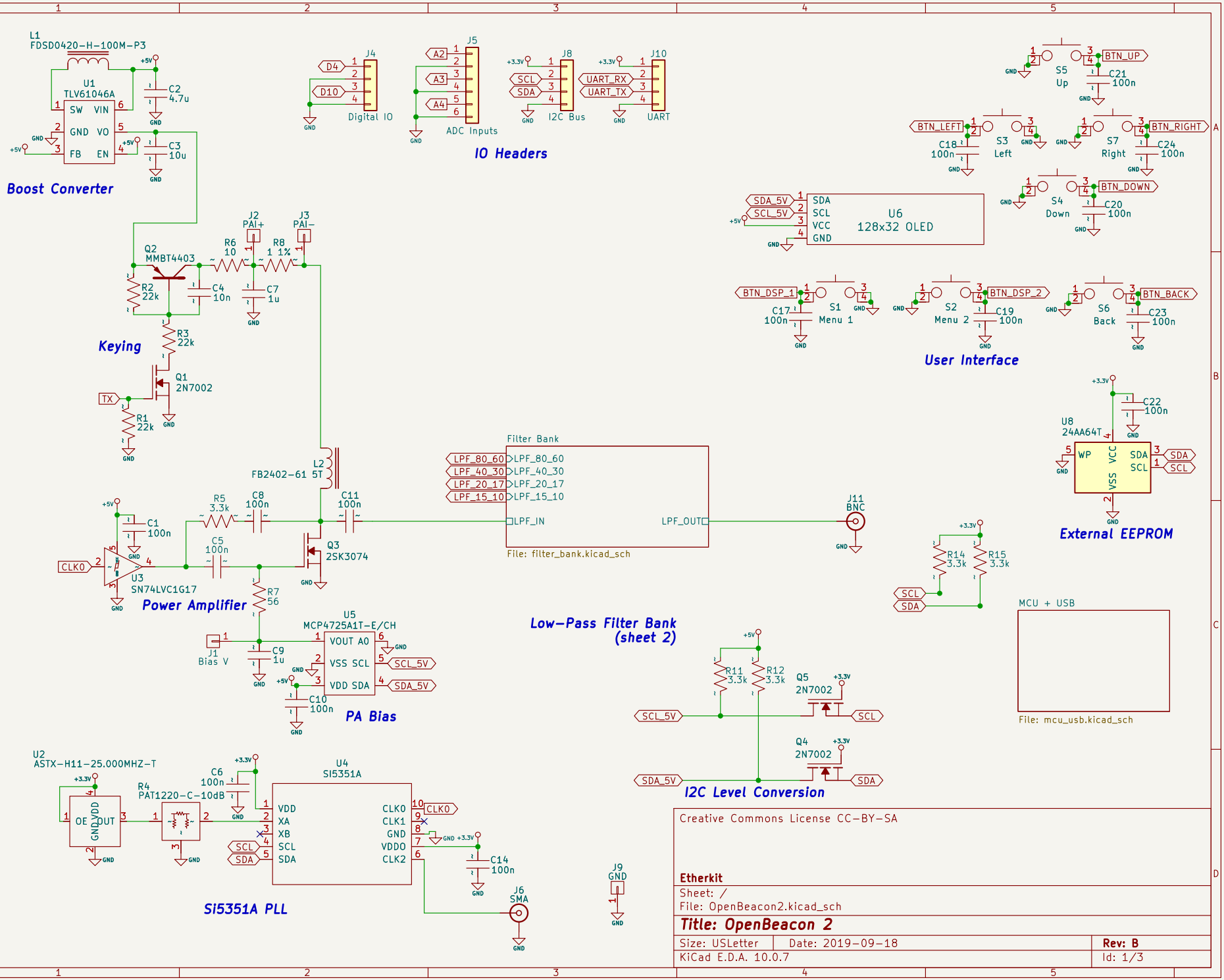
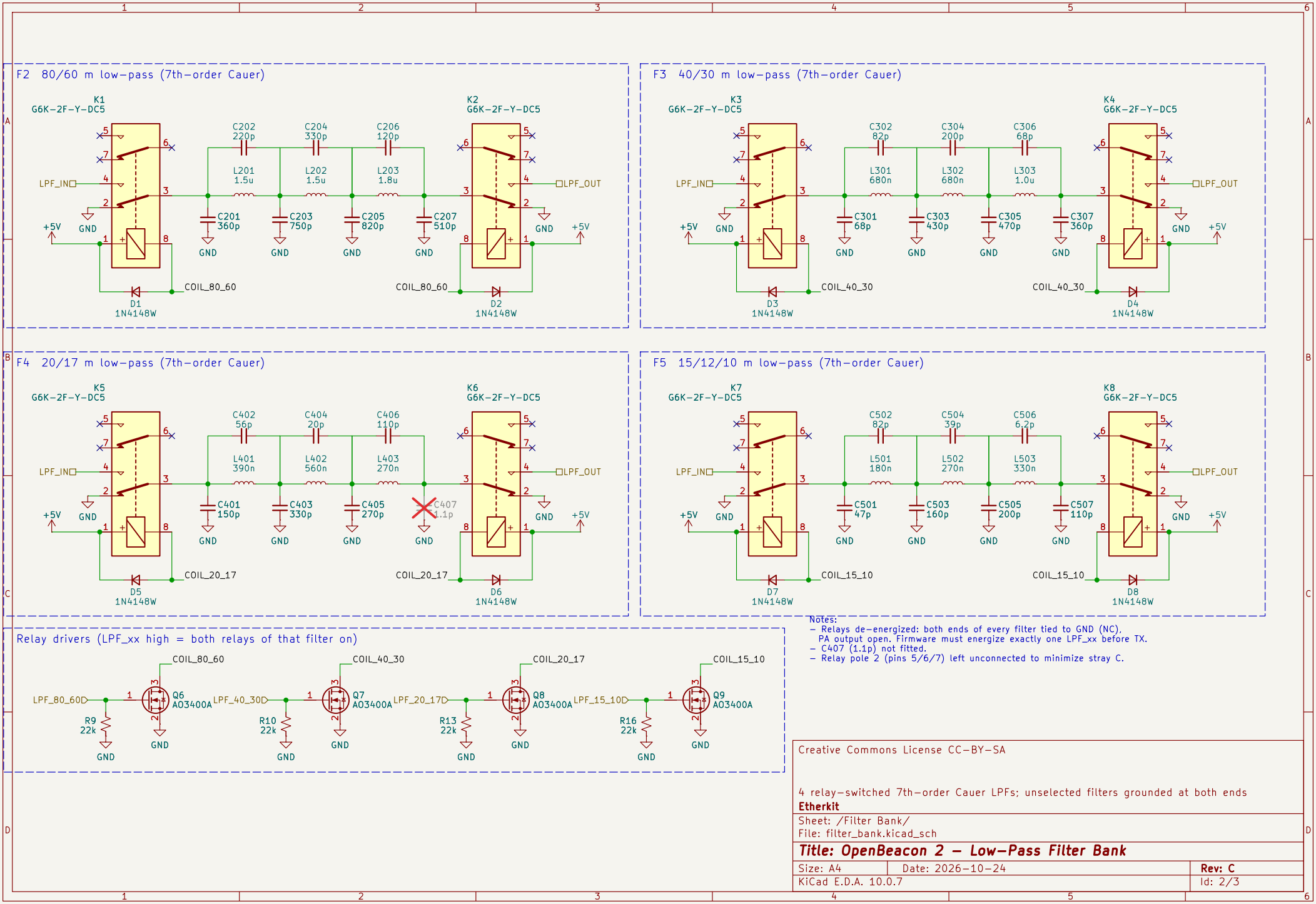
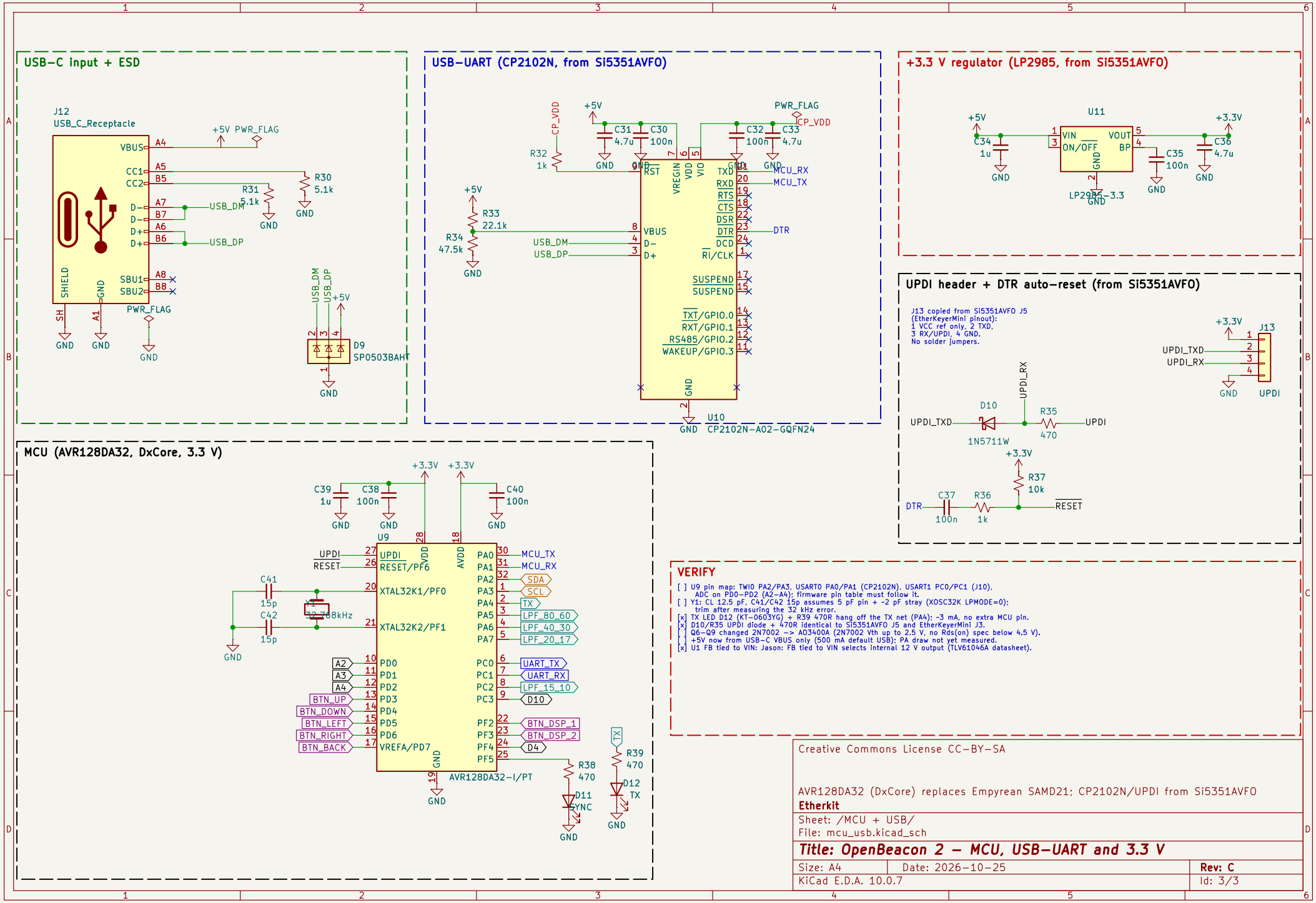

Boost Converter
IO Headers
Digital IO
ADC Inputs
I2C Bus
UART
User Interface
Keying
Power Amplifier
PA Bias
Low-Pass Filter Bank
(sheet 2)
I2C Level Conversion
External EEPROM
SI5351A PLL
MCU + USB
File: filter_bank.kicad_sch
File: mcu_usb.kicad_sch
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Etherkit
Sheet: /
File: OpenBeacon2.kicad_sch
Title: OpenBeacon 2
Size: USLetter
Date: 2019-09-18
Rev: B
KiCad E.D.A. 10.0.7
Id: 1/3


F2 80/60 m low-pass (7th-order Cauer)
K1
G6K-2F-Y-DC5
K2
G6K-2F-Y-DC5
C202 220p
C204 330p
C206 120p
L201 1.5u
L202 1.5u
L203 1.8u
C201 360p
C203 750p
C205 820p
C207 510p
LPF_IN
LPF_OUT
+5V
GND
COIL_80_60
D1 1N4148W
D2 1N4148W
F3 40/30 m low-pass (7th-order Cauer)
K3
G6K-2F-Y-DC5
K4
G6K-2F-Y-DC5
C302 82p
C304 200p
C306 68p
L301 680n
L302 680n
L303 1.0u
C301 68p
C303 430p
C305 470p
C307 360p
COIL_40_30
D3 1N4148W
D4 1N4148W
F4 20/17 m low-pass (7th-order Cauer)
K5
G6K-2F-Y-DC5
K6
G6K-2F-Y-DC5
C402 56p
C404 20p
C406 110p
L401 390n
L402 560n
L403 270n
C401 150p
C403 330p
C405 270p
C407 1.1p
COIL_20_17
D5 1N4148W
D6 1N4148W
F5 15/12/10 m low-pass (7th-order Cauer)
K7
G6K-2F-Y-DC5
K8
G6K-2F-Y-DC5
C502 82p
C504 39p
C506 6.2p
L501 180n
L502 270n
L503 330n
C501 47p
C503 160p
C505 200p
C507 110p
COIL_15_10
D7 1N4148W
D8 1N4148W
Relay drivers (LPF_xx high = both relays of that filter on)
LPF_80_60
LPF_40_30
LPF_20_17
LPF_15_10
Q6 AO3400A
Q7 AO3400A
Q8 AO3400A
Q9 AO3400A
R9 22k
R10 22k
R13 22k
R16 22k
Notes:
– Relays de-energized: both ends of every filter tied to GND (NC).
PA output open. Firmware must energize exactly one LPF_xx before TX.
– C407 (1.1p) not fitted.
– Relay pole 2 (pins 5/6/7) left unconnected to minimize stray C.
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4 relay-switched 7th-order Cauer LPFs; unselected filters grounded at both ends
Etherkit
Sheet: /Filter Bank/
File: filter_bank.kicad_sch
Title: OpenBeacon 2 – Low-Pass Filter Bank
Size: A4
Date: 2026-10-24
Rev: C
KiCad E.D.A. 10.0.7
Id: 2/3


USB-C input + ESD
J12
USB_C_Receptacle
VBUS
CC1
CC2
D-
D-
D+
D+
SHIELD
GND
SBU1
SBU2
A4
A5
B5
A7
B7
A6
B6
A8
B8
SH
A1
+5V PWR_FLAG
R30
5.1k
R31
5.1k
USB_DM
USB_DP
PWR_FLAG
GND
USB_DM
USB_DP
+5V
D9
SP0503BAHT
USB-UART (CP2102N, from Si5351AVFO)
CP_VDD
+5V
C31
4.7u
C30
100n
PWR_FLAG
C32
100n
C33
4.7u
R32
1k
RST
VREGIN
VDD
VIO
TXD
RXD
RTS
CTS
DSR
DTR
DCD
RI/CLK
SUSPEND
SUSPEND
TXT/GPIO.0
RXT/GPIO.1
RS485/GPIO.2
WAKEUP/GPIO.3
MCU_RX
MCU_TX
DTR
R33
22.1k
R34
47.5k
VBUS
D-
D+
U10
CP2102N-A02-GQFN24
+3.3 V regulator (LP2985, from Si5351AVFO)
U11
C34
1u
VIN
ON/OFF
GND
VOUT
BP
C35
100n
C36
4.7u
+3.3V
LP2985-3.3
UPDI header + DTR auto-reset (from Si5351AVFO)
J13 copied from Si5351AVFO J5
(EtherKeyerMini pinout):
1 VCC ref only, 2 TXD,
3 RX/UPDI, 4 GND.
No solder jumpers.
J13
UPDI_TXD
UPDI_RX
UPDI
D10
1N5711W
UPDI_RX
R35
470
R37
10k
C37
100n
R36
1k
DTR
RESET
MCU (AVR128DA32, DxCore, 3.3 V)
C39
1u
C38
100n
C40
100n
U9
UPDI
RESET/PF6
XTAL32K1/PF0
XTAL32K2/PF1
PD0
PD1
PD2
PD3
PD4
PD5
PD6
VREFA/PD7
VDD
AVDD
GND
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PC0
PC1
PC2
PC3
PF2
PF3
PF4
PF5
AVR128DA32-I/PT
C41
15p
C42
15p
32.768kHz
A2
A3
A4
BTN_UP
BTN_DOWN
BTN_LEFT
BTN_RIGHT
BTN_BACK
MCU_TX
MCU_RX
SDA
SCL
TX
LPF_80_60
LPF_40_30
LPF_20_17
UART_TX
UART_RX
LPF_15_10
D10
BTN_DSP_1
BTN_DSP_2
D4
R38
470
D11
SYNC
R39
470
D12
TX
VERIFY
[ ] U9 pin map: TWI0 PA2/PA3, USART0 PA0/PA1 (CP2102N), USART1 PC0/PC1 (J10),
ADC on PD0–PD2 (A2–A4); firmware pin table must follow it.
[ ] Y1: CL 12.5 pF, C41/C42 15p assumes 5 pF pin + ~2 pF stray (XOSC32K LPMODE=0);
trim after measuring the 32 kHz error.
[x] TX LED D12 (KT-0603YG) + R39 470R hang off the TX net (PA4): ~3 mA, no extra MCU pin.
[x] D10/R35 UPDI diode + 470R identical to Si5351AVFO J5 and EtherKeyerMini J3.
[ ] Q6–Q9 changed 2N7002 -> AO3400A (2N7002 Vth up to 2.5 V, no Rds(on) spec below 4.5 V).
[ ] +5V now from USB-C VBUS only (500 mA default USB); PA draw not yet measured.
[x] U1 FB tied to VIN: Jason: FB tied to VIN selects internal 12 V output (TLV61046A datasheet).
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AVR128DA32 (DxCore) replaces Empyrean SAMD21; CP2102N/UPDI from Si5351AVFO
Etherkit
Sheet: /MCU + USB/
File: mcu_usb.kicad_sch
Title: OpenBeacon 2 – MCU, USB-UART and 3.3 V
Size: A4
Date: 2026-10-25
Rev: C
KiCad E.D.A. 10.0.7
Id: 3/3